

STATS 200A Lecture 2

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$$\mathcal{F}_1 \cap \mathcal{F}_2 = \{\emptyset, \Omega\} = \mathcal{F}_0$$

$\mathcal{F}_1 \cap \mathcal{F}_2$ not necessarily a σ -algebra, however, $\sigma(F_1 \cap F_2)$ is
 $\sigma(\mathcal{F}_1 \cap \mathcal{F}_2) = \sigma(\underbrace{\{A \cap B : A \in \mathcal{F}_1, B \in \mathcal{F}_2\}}_{\text{is a } \pi\text{-system (closed under intersection)}})$

Example 2.1

$$\Omega = \{1, 2, 3, 4, 5, 6\}, \mathcal{F} = 2^\Omega, \mu(A) = \frac{|A|}{6}, A \in \mathcal{F}$$

Solution. $\mu(\{1, 3, 5\}) = \frac{3}{6} = \frac{1}{2}$ ■

Definition 2.2 (measure)

A *measure* on $(\Omega, \mathcal{F})$ is $\mu : \mathcal{F} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$ and

$$\mu\left(\bigcup_n A_n\right) = \sum_n \mu(A_n) \quad \text{for pairwise disjoint } A_1, A_2, \dots \in \mathcal{F}.$$

$\mu(\Omega) = 1$: *Probability* measure $\mathbb{P}$; $(\Omega, \mathcal{F}, \mathbb{P})$ a *probability space*
 $\mu(\Omega) < \infty$: *finite*. $\Omega = \bigcup_n \Omega_n$ with $\mu(\Omega_n) < \infty$: σ -finite
 Ω is covered by a countable union

Lesbesgue measure: $(\mathbb{R}, B(\mathbb{R}), \lambda) : \lambda((a, b]) = b - a$

$$\bigcup_{n \geq 1} (-n, n] = \mathbb{R}$$

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$$

Q: $(\mathbb{R}, B(\mathbb{R}), \mu)$ where μ is a probability measure, cdf is an example of μ

$$\mu((a, b]) = F(b) - F(a) \quad \mu((-n, n]) = F(n) - F(-n) \xrightarrow{n \rightarrow \infty} 1$$

$$\mu((1, 2] \cup (2, 3]) = \mu((1, 2]) + \mu((2, 3]) = [F(2) - F(1)] + [F(3) - F(2)] = F(3) - F(1)$$

$$B(\mathbb{R}) = \sigma(\{(a, b] : a, b \in \mathbb{R}\})$$

Remark 2.3 (Common Measures)

Counting Measure: $\mu(A) = |A|$ if finite, otherwise ∞

Point mass at x : $\delta_x(A) = \mathbf{1}\{x \in A\}$

Discrete measure: $\mu(A) = \sum_i p_i \delta_{x_i}(A), p_i \geq 0$

Measure on a finite σ -algebra: A vector on the atoms: $\mu(A) = \sum_{A_i \subseteq A} \mu(A_i)$

Lesbesgue measure λ on $(\mathbb{R}, B(\mathbb{R}))$: $\lambda((a, b]) = b - a$

$$\mu = \frac{1}{5}\delta_3 + \frac{4}{5}\delta_1 \text{ "mixture of measures"}$$

$$\mu = \alpha\mu_1 + (1 - \alpha)\mu_2, \alpha \in [0, 1]$$

$$\mu = \sum_{i=1}^n \alpha_i \mu_i, \alpha_i \geq 0, \sum_{i=1}^n \alpha_i = 1$$

$$A_n \uparrow A : A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots, A = \bigcup_{n \geq 1} A_n$$

$$A_n \downarrow A : A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots, A = \bigcap_{n \geq 1} A_n$$